DSFB Computer Graphics Reviewer Bundle

This PDF is generated from a single timestamped run directory so reviewers can archive the bounded synthetic artifact without depending on the notebook session state.

"The experiment is intended to demonstrate behavioral differences rather than establish optimal performance."

What This Bundle Contains

Demo A figures, key numeric metrics, an interpretation grounded in the actual run outputs, and the main limitations of the artifact.

The bundle is intended to support technical review, transition conversations, and reproducible replay rather than optimality claims or deployment claims.

Run directory: output-dsfb-computer-graphics-20260323-011723

Key Metrics

In this bounded synthetic setting, DSFB reduced ghost persistence duration from 12 to 0 frames relative to the fixed-alpha baseline.

Against the residual-threshold baseline, DSFB reduced ghost persistence duration from 4 to 0 frames.

Metric	Fixed-alpha	Residual-threshold	DSFB
Ghost persistence frames	12	4	0
Peak ROI error	0.39133	0.12452	0.06475
Cumulative ROI error	2.55778	0.70179	0.21577
Average overall MAE	0.01228	0.00373	0.00220

Event Timing

Reveal frame: 6

Trust-drop frame: 6

Trust-minimum frame: 6

Residual-baseline response frame: 6

Trust/error correlation at reveal: 0.9071

Figure 1. System Diagram

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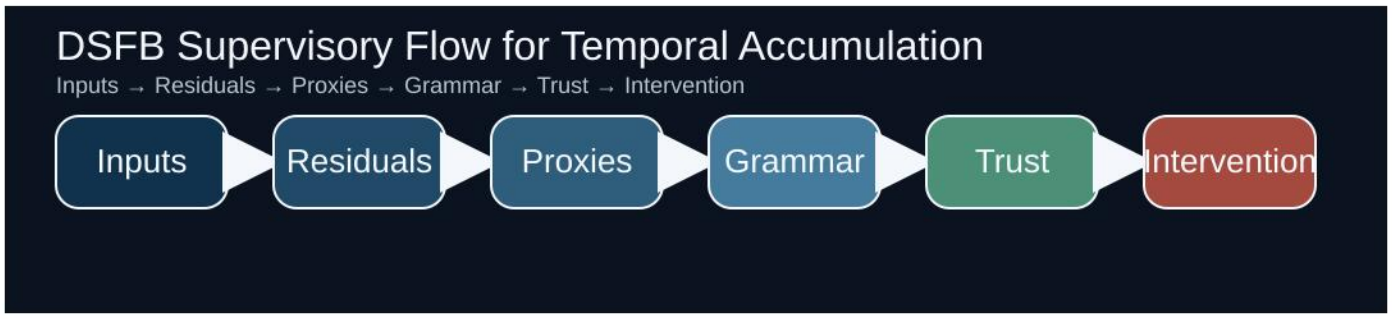

Inputs, residuals, proxies, simplified grammar, trust, and intervention as used in the current crate.

Figure 2. Trust Map

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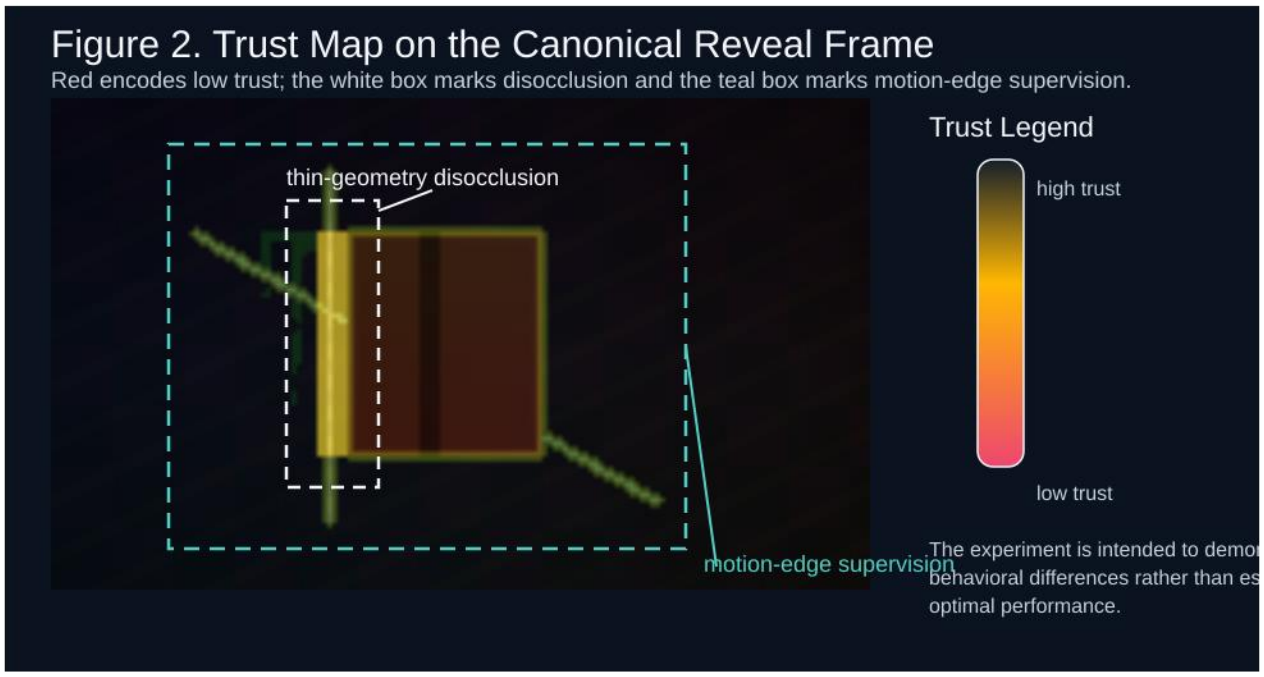

Low trust concentrates around the reveal event and motion-edge supervision region on the actual demo frame.

Figure 3. Before / After

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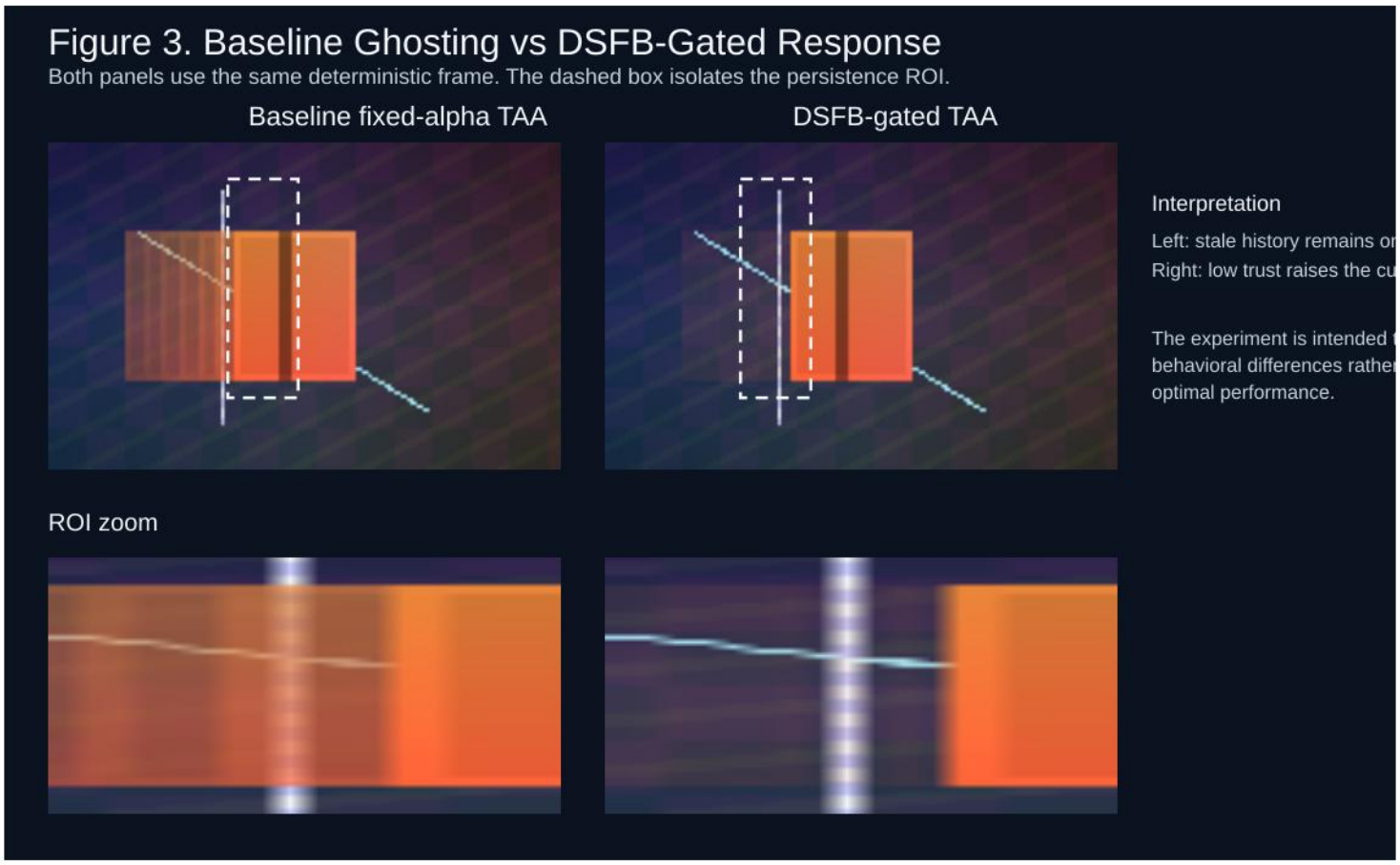

Fixed-alpha baseline versus DSFB on the same comparison frame and persistence ROI.

Figure 4. Trust vs Error

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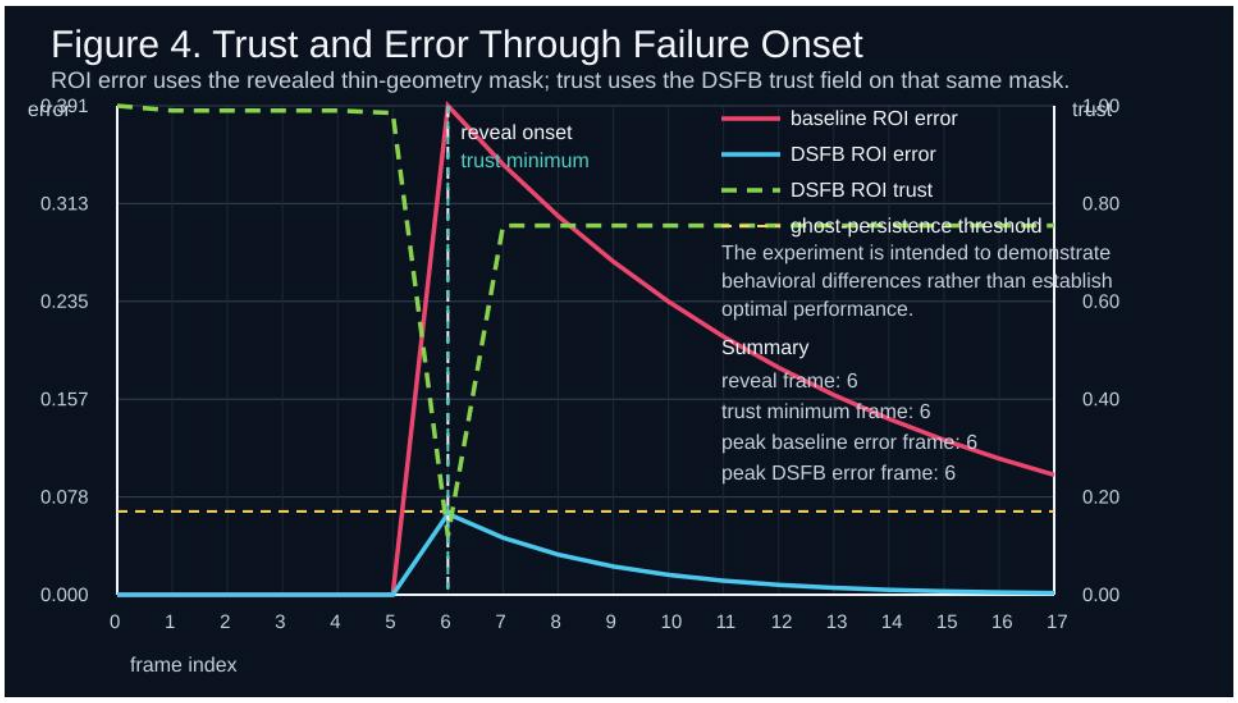


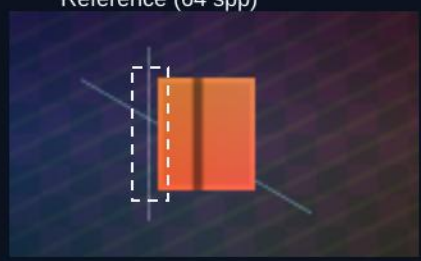
Frame index on the x-axis, ROI error on the left y-axis, and DSFB ROI trust on the right y-axis.

Demo B. Fixed-Budget Adaptive Sampling

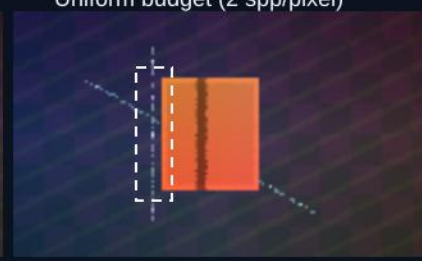
Demo B. Fixed-Budget Adaptive Sampling on the Reveal Frame

Top row: reference, uniform allocation, DSFB-guided allocation. Bottom row: uniform error, DSFB error, guided sample density.

Reference (64 spp)



Uniform budget (2 spp/pixel)



DSFB-guided fixed budget

Uniform absolute error

DSFB-guided absolute error

Guided samples per pixel

Measured outcome

Uniform MAE: 0.00289
Guided MAE: 0.00250
Uniform ROI MAE: 0.17348
Guided ROI MAE: 0.03714
ROI mean spp: uniform 2, guided 12.00
Guided max spp: 12

The experiment is intended to demonstrate behavioral differences rather than establish optimal performance.

At equal total sample budget, the guided allocation concentrates work in the low-trust reveal ROI. Uniform ROI MAE is 0.17348; guided ROI MAE is 0.03714.

Limitations and Scope

This remains a bounded synthetic artifact. It does not claim production-optimal TAA, measured GPU timing wins, field readiness, or universal superiority over commercial reconstruction systems.

PDF and ZIP export exist to make external review easier: a reviewer can archive the exact bundle from one run directory and inspect the same figures and metrics later without rerunning the notebook.

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